

Patient-robust discovery of combinatorial cell-surface targets in prostate cancer

AND-gate and NOT-gate marker pairs from single-cell data, benchmarked against PSMA-PSCA

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Abstract Single-antigen therapies against solid tumors are limited by on-target off-tumor toxicity, because few surface antigens are truly absent from all healthy tissue. Combinatorial (logic-gated) targeting requires two conditions before a cell is engaged, which can recover specificity. We scan a curated cell-surface panel in single-cell data for pairs that discriminate prostate cancer cells from healthy human cell types, under AND (both markers on the tumor) and NOT (an activator spared by a blocker on healthy cells) logic. Pairs are scored per patient and summarized across patients, so no high cell-count patient drives a result. The preclinically validated PSMA-PSCA balanced-signalling pair is recovered as a positive control: each marker alone carries a large extra-prostatic transcript liability, and the AND gate collapses it. We report a Pareto frontier of surface-accessible pairs with higher per-patient malignant co-detection than PSMA-PSCA at comparable predicted normal-tissue liability, with Human Protein Atlas evidence supporting membrane-localization plausibility. These are transcript-level, hypothesis-generating rankings, not evidence of therapeutic safety.

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1 Background

A major challenge for antigen-directed therapies in solid tumors is achieving tumor coverage without clinically important on-target, off-tumor activity. A single tumor antigen is rarely absent from every healthy tissue, so single-target CAR-T, T-cell engagers, and antibody-drug conjugates carry this risk. Combinatorial, logic-gated targeting requires two conditions before a cell is engaged and can recover the missing specificity [1]–[3]. Prostate adenocarcinoma is the natural test bed: the PSMA (FOLH1) x PSCA pair is a preclinically validated AND-gate split-signal CAR [1], so a systematic scan of a curated panel can be checked against a known answer, and STEAP1 anchors a second axis with a STEAP1 x CD3 T-cell engager (xaluritamig) in trials [4]. That trial establishes STEAP1 as druggable in prostate cancer; it does not validate a PSMA x STEAP1 gate.

Computational logic-gated antigen discovery is not itself new: pan-cancer single-cell pair scans [5], a breast-cancer AND/OR/NOT circuit search [6], and a single-cell AND / AND-NOT sarcoma target scan with a co-expression error model [7] precede this work. Our contribution is the combination of (i) per-patient scoring rather than pooled cells, (ii) a matched benign-prostate control that tests tumor specificity in the same samples, (iii) an assay-matched, donor-robust healthy reference so normal-tissue liabilities are comparable across platforms and not driven by a single donor, and (iv) recovery of the PSMA-PSCA benchmark before any new pair is nominated. The nominated target pair is not itself novel: a dual PSMA/STEAP1 antibody-drug conjugate (ABBV-969) is in phase 1 [8], though as an OR-like agent that binds either antigen rather than a strict AND gate.

2 Data

- **Tumor discovery cohort** — 68,322 cells from **24 patients** with localised, hormone therapy-naive prostate adenocarcinoma [9]. Malignant identity uses the authors’ copy-number and signature-based annotation: **3,590 malignant cells** (scDblFinder singlets; see Methods), with **16,249 adjacent-benign / normal prostate epithelial cells** retained as a matched same-patient control.
- **Replication cohort** — HuPSA [10], an integrated prostate-cancer atlas (~369,000 cells, 6 studies, 10x) spanning normal, localised, castration-resistant, metastatic and neuroendocrine disease, used to test whether the nominated pair generalises beyond localised disease (see Replication).
- **Healthy reference** — Tabula Sapiens 2.0 [11], [12], pulled through the CELLxGENE Census (release 2025-01-30) and restricted to the **10x 3’ v3** assay to match the tumor cohort chemistry, since a raw-count positivity threshold is not comparable across assays: **1,025,717 cells**, **173 cell types** across **65 tissue labels** (about 30 organs); **442 donor-replicated tissue-by-cell-type populations** (at least 2 donors each) form the safety denominator.

- **Protein evidence** — Human Protein Atlas subcellular localization and normal-tissue RNA [13], used to qualify surface accessibility and cross-check normal-tissue liabilities.
- **Panel** — 29 curated cell-surface markers scanned (see Methods); secreted markers (KLK3/PSA, KLK2, ACP) are kept only as comparators.

3 Methods

- **Doublet removal (primary input).** scDblFinder, run per patient in a Bioconductor container, flagged 4.8% of tumor cells as doublets (5.6% of malignant cells); these are removed before scoring, so a two-cells-as-one capture cannot fake AND co-expression (Rule 8). Re-scoring on the 65,040 retained single cells moves the key pairs' AND coverage by at most 1.0 percentage points, so the finding is not a doublet artifact.

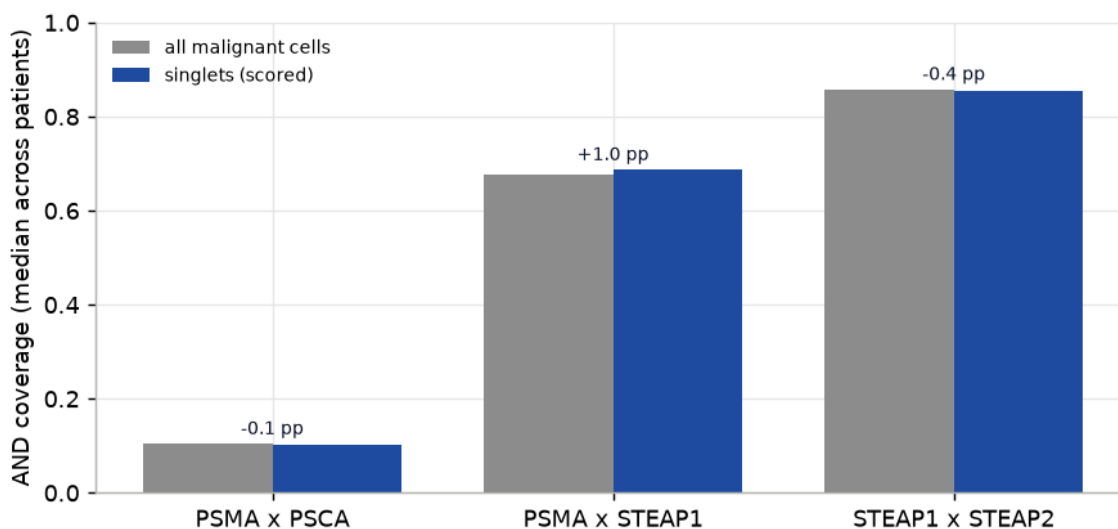


Figure 1: Doublet robustness. Median per-patient AND coverage of the key pairs computed on all malignant cells (grey) versus the scDblFinder singlets actually used for scoring (blue). The bars are near-identical and the change (annotated, percentage points) is at most one point, so the co-expression signal is not an artifact of two cells captured as one; removing doublets nudges the nominated pairs slightly upward.

- **Positivity** is a per-cell raw count of at least 1 (single-cell detection, not a threshold on a pooled or averaged expression value); the threshold is swept for sensitivity (below).
- **Per-patient scoring (never pooled).** For markers A, B and each patient's malignant cells, AND coverage is $P(A+ \text{ and } B+ \mid \text{malignant})$ computed within that patient, then summarized across patients by the lower decile (Q0.10, the robustness floor) and the median. A patient with thousands of cells cannot dominate a pair's score.
- **Three references per pair.** (1) malignant coverage; (2) **matched benign prostate** from the same cohort, so a pair expressed across all prostate cells is penalised, not just one specific to the tumor;

- (3) **healthy liability** as the worst-case co-positive fraction over Tabula Sapiens populations. The healthy side is assay-matched to the tumor (10x 3' v3 only) and made **donor-robust**: within each tissue-by-cell-type population the fraction is computed per donor and summarized by the median across donors, keeping only populations with at least two donors, so no single donor or tiny population sets a worst case. Every worst-liability call carries its supporting donor and cell counts. Liabilities are reported both across all organs (the stricter statistic, retained in the tables) and **excluding the prostate** (normal-prostate expression may be clinically tolerable in selected advanced-disease contexts, whereas every other organ is dose-limiting); the figures use the extra-prostatic statistic.
- **Label-leakage control.** The cohort's malignant label is built from the Liu and Wallace prostate-cancer expression signatures plus CopyKAT aneuploidy. HPN is a member of both signatures and EPCAM of the Liu signature, so any pair containing HPN or EPCAM is confounded: the gene helps define the label it then scores against. Such pairs are flagged, held off the Pareto frontier, and not nominated; the nominated PSMA x STEAP1 lead and the PSMA-PSCA control contain no signature gene. A full fix recomputes the labels with all 29 screened genes removed from the signatures (future work).
 - **Gate algebra.** For each group a single matrix product over the float32 panel positivity matrix yields AND, NOT (A+ and B-) and OR counts for every pair at once; co-positive counts stay far below the float32 exact-integer limit, so there is no overflow, and NOT gates are directed (all 812 activator-blocker orientations are scored). These Boolean gates are screening abstractions of candidate receptor logic; they do not model antigen-density thresholds, residual single-arm activity, or the architecture-specific signalling of a balanced split-signal CAR, synNotch, or Tmod circuit.
 - **Ranking.** Pairs are placed on a Pareto frontier of per-patient coverage floor (higher better) against worst extra-prostatic liability (lower better). Only surface-accessible markers are nom-inatable.
 - **Reproducibility.** Every number below is read from a committed table in `results/tables/`; the report redraws its figures from those tables with no network or model call. The healthy pull uses the Census raw-count layer with `is_primary_data == True` (deduplicated cells), release 2025-01-30, and confirms the panel features are present in the query (ACPP was the one panel gene absent).

4 Results

4.1 Single surface antigens each carry a large normal-tissue liability

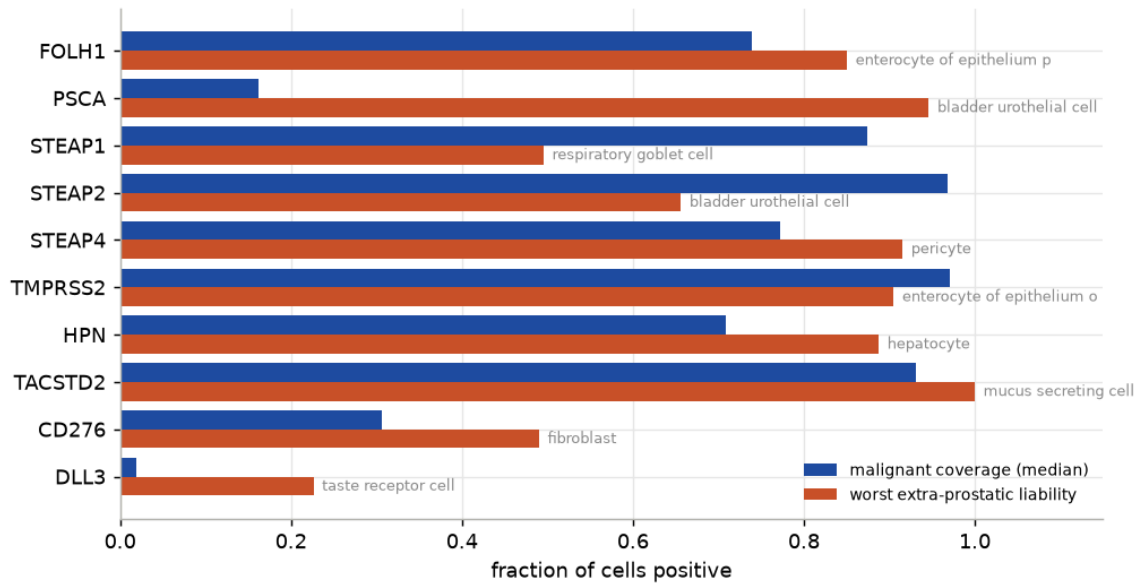


Figure 2: Single surface antigens: per-patient malignant coverage (median) against the worst-case extra-prostatic normal-tissue co-detection. Every strong single antigen also carries a large normal-tissue transcript liability; the recovered worst tissues are concordant with known biology (FOLH1/PSMA to duodenum, PSCA to bladder urothelium).

- Each strong single antigen is broadly positive on the tumor but also on at least one healthy cell type: the scan independently recovers PSMA’s duodenal and PSCA’s bladder-urothelial expression, a qualitative sanity check that the recovered tissues match known biology, not a claim that the fractions are quantitatively exact.

4.2 The AND gate recovers PSMA-PSCA and collapses its single-agent liabilities

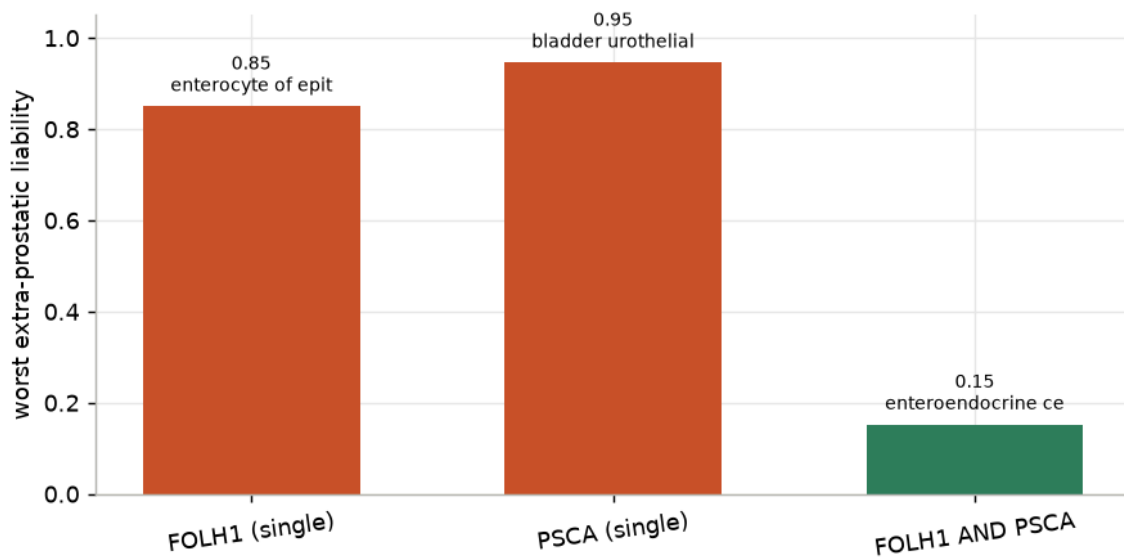


Figure 3: Positive control. PSMA (FOLH1) and PSCA are each broadly detected off the prostate (duodenum, bladder), but requiring both at once (AND) collapses the worst extra-prostatic co-detection roughly sixfold, recapitulating the direction of the preclinical rationale for the split-signal PSMA x PSCA CAR.

- **Recovered.** PSMA alone reaches **0.85** co-detection in duodenal enterocytes and PSCA alone **0.95** in bladder urothelium; the **PSMA and PSCA** gate drops the worst extra-prostatic co-detection to **0.15** (a reduction of 0.79), the expected direction of the preclinical benchmark.
- **Against a random baseline.** 57% of the sampled surface pairs (drawn from the same curated panel) score worse than PSMA-PSCA on extra-prostatic selectivity, defined as median malignant co-detection minus worst extra-prostatic liability. This is a weak signal: a pair can also look clean by carrying little tumor coverage, so it is a sanity check, not proof of specificity.
- Its low malignant co-detection (0.10 median) is consistent with limited PSCA transcript detection in this dataset; protein-level measurement would be needed to separate technical non-detection from biological heterogeneity. It motivates looking for higher-coverage pairs of comparable liability.

4.3 A Pareto frontier of surface pairs with higher coverage than PSMA-PSCA

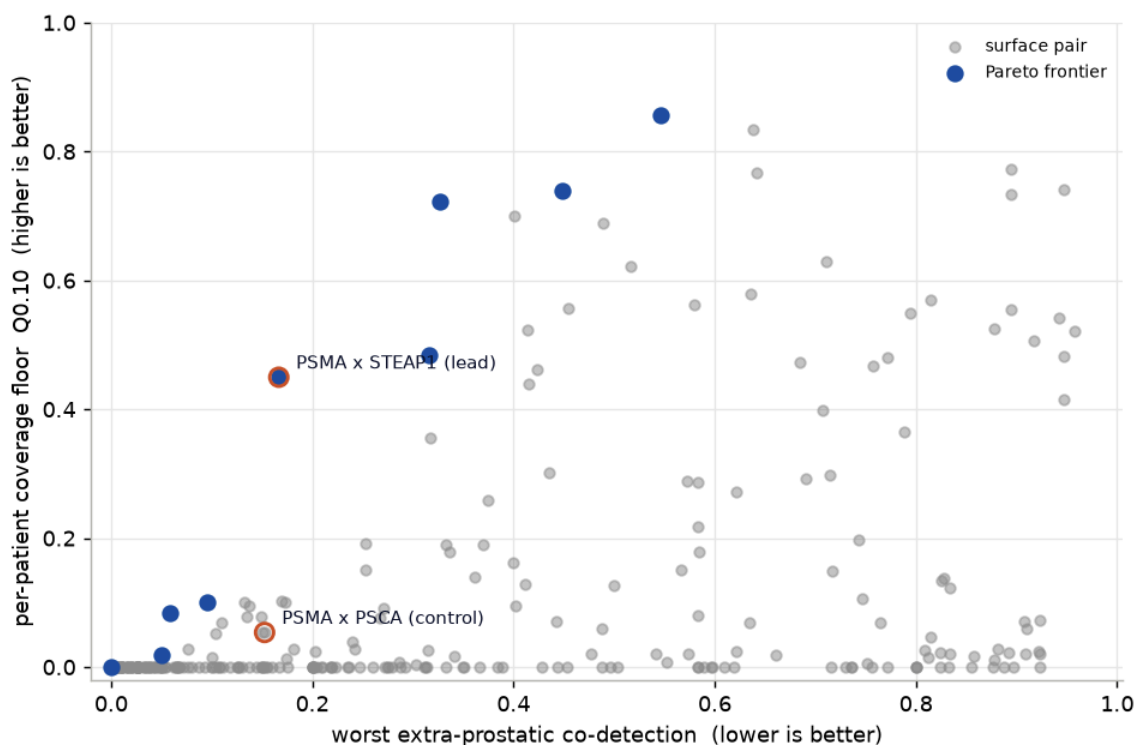


Figure 4: Surface-accessible AND pairs (label-leakage pairs excluded): per-patient coverage floor (Q0.10) against worst extra-prostatic co-detection. Upper-left is better. Frontier pairs are filled; the PSMA-PSCA control and the nominated PSMA x STEAP1 lead are labelled. Excluded as non-nominatable: secreted markers, intracellular protein, and any pair containing a malignant-label signature gene (HPN, EPCAM).

One surface-accessible pair is nominated. It is clean of label leakage (neither marker is a malignant-label signature gene) and, once the leakage-confounded HPN pairs are removed, it is Pareto-non-dominated among the surface-accessible candidates.

- **PSMA x STEAP1 — nominated lead.** Both antigens have clinical-grade binders. Per-patient coverage floor **Q0.10 = 0.45**, median **0.69** (**6.7x** the PSMA-PSCA median). Its worst extra-prostatic co-detection is **0.17** in mucus secreting cell (5 donors), versus 0.85 / 0.50 for the single antigens and 0.15 for PSMA-PSCA. Whether that residual liability is clinically acceptable cannot be judged from transcript data. Median coverage 0.69 and floor 0.45 are not obviously adequate for monotherapy; a sizeable tumor fraction can remain double-negative or single-positive.
- **Higher in malignant than adjacent-benign prostate.** The pair shows more co-detection in annotated malignant than in matched benign prostate epithelium (malignant-minus-benign 0.60).
- **Coverage is confined to AR-driven adenocarcinoma.** Replication in the independent HuPSA cohort holds in AR+ adenocarcinoma but collapses in castration-resistant, metastatic and neuroendocrine disease (see Replication), so this coverage does not extend to the advanced-disease population.

- **The pair is not novel, and AND is not the only architecture.** A dual PSMA/STEAP1 antibody-drug conjugate (ABBV-969) is in phase 1 [8], but as a single molecule binding either antigen it is OR-like, not a strict AND gate. Our contribution is the per-patient, benchmark-anchored AND framing, not the target pair.
- **HPN pairs are excluded as confounded.** STEAP1 x HPN and PSMA x HPN scored well, but HPN is a member of the Liu and Wallace signatures used to build the malignant-cell label, so their coverage is inflated by label leakage (see Methods); they are not nominated pending re-computation of the labels without the screened genes.

Surface-accessible AND candidates (top 15 of 276 clean targetable pairs; a further 49 pairs containing a malignant-label signature gene, HPN or EPCAM, are excluded as label-leakage confounded). Ranked by extra-prostatic selectivity: median malignant co-detection minus worst extra-prostatic co-detection. * marks the nominated lead; frontier marks Pareto-non-dominated pairs; donors supports the worst-liability population. Hypothesis-generating rankings over transcript co-detection, not validated targets.

pair		Q0.10	me- dian	worst xp	worst cell type		donors	mal- ben	frontier
STEAP1 STEAP2	x	0.72	0.85	0.33	respiratory cell	goblet	4	0.6	yes
PSMA STEAP1 *	x	0.45	0.69	0.17	mucus cell	secreting	5	0.6	yes
STEAP1 TMPRSS2	x	0.7	0.84	0.4	bladder cell	urothelial	6	0.6	
PSMA STEAP2	x	0.48	0.73	0.32	mucus cell	secreting	5	0.64	yes
STEAP1 CD46	x	0.74	0.85	0.45	bladder cell	urothelial	6	0.52	yes
STEAP2 TMPRSS2	x	0.86	0.93	0.55	bladder cell	urothelial	6	0.64	yes
STEAP1 TACSTD2	x	0.69	0.84	0.49	respiratory cell	goblet	4	0.49	
STEAP1 STEAP4	x	0.52	0.74	0.41	respiratory cell	goblet	4	0.57	
PSMA TACSTD2	x	0.44	0.71	0.41	mucus cell	secreting	5	0.62	
STEAP1 CDH1	x	0.56	0.74	0.45	respiratory cell	goblet	4	0.5	

pair		Q0.10	me- dian	worst xp	worst cell type	donors	mal- ben	frontier
STEAP1 TMEFF2	x	0.08	0.34	0.06	epithelial cell	2	0.33	yes
STEAP2 CD46	x	0.83	0.91	0.64	bladder urothelial cell	6	0.57	
PSMA STEAP4	x	0.36	0.59	0.32	mucus secreting cell	5	0.52	
STEAP2 TMEFF2	x	0.1	0.35	0.1	retinal bipolar neuron	3	0.34	yes
STEAP2 TACSTD2	x	0.77	0.9	0.64	bladder urothelial cell	6	0.54	

4.4 Robustness: per patient and across positivity thresholds

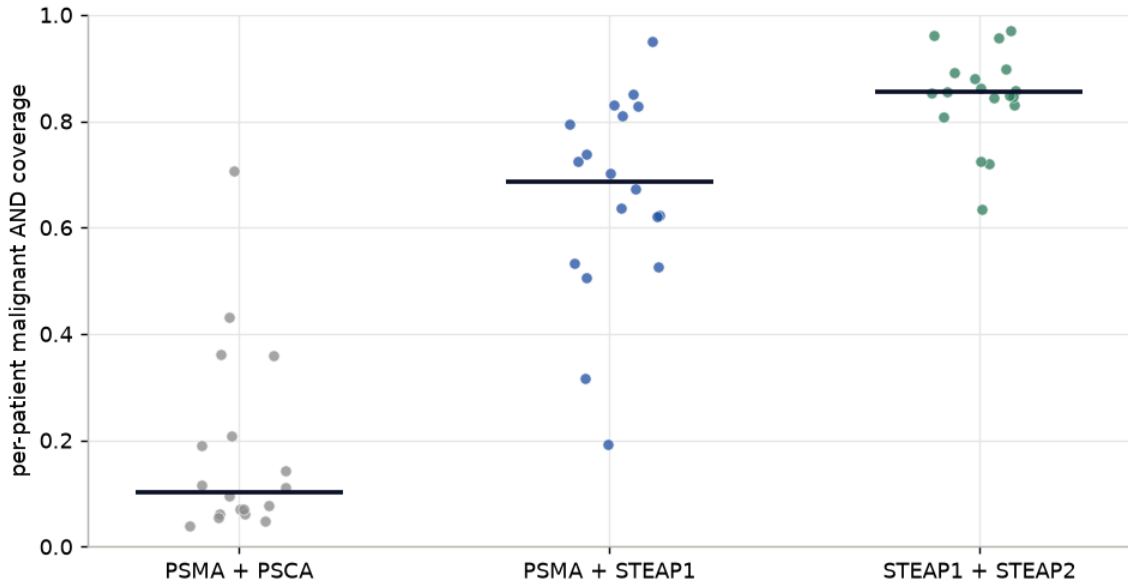


Figure 5: Per-patient AND co-detection of the malignant compartment for the positive control, the nominated lead, and a clean high-coverage comparator. Each point is one patient; the bar is the median. PSMA x STEAP1 (lead) and STEAP1 x STEAP2 (clean, not nominated) hold a per-patient coverage floor (Q0.10) above 0.44 across scored patients, while PSMA x PSCA is uniformly low (limited PSCA transcript detection).

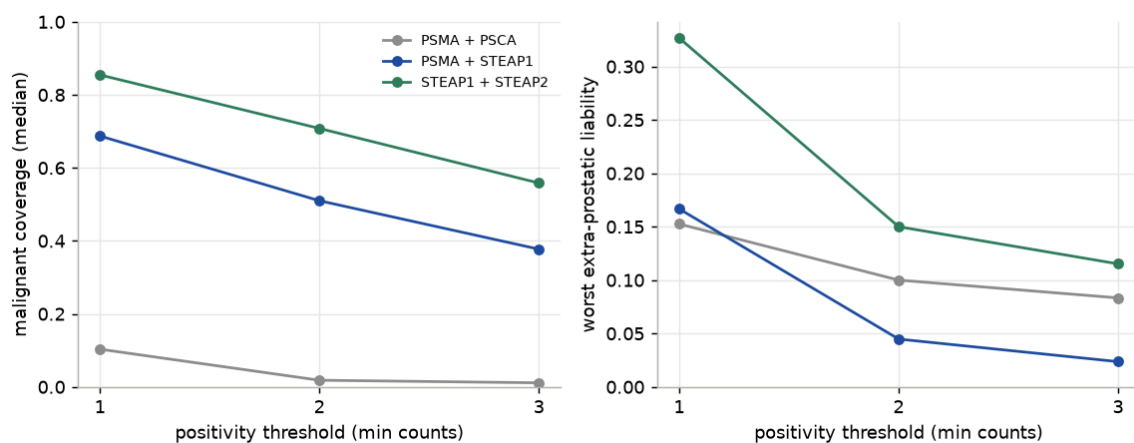


Figure 6: Positivity-threshold sensitivity. As the per-cell call is tightened from ≥ 1 to ≥ 3 counts, malignant co-detection falls modestly while the worst extra-prostatic co-detection falls too: the candidates stay well covered and their normal-tissue liability drops, whereas PSMA x PSCA loses coverage. Positivity is a per-cell detection threshold, swept here, not a pooled-mean call.

- Both candidates survive threshold tightening: malignant co-detection decreases gradually while the worst extra-prostatic co-detection falls, so the selectivity ranking is not an artifact of the loosest positivity call.

Residual off-tumor liabilities of PSMA x STEAP1 are named, not hidden. The highest extra-prostatic healthy populations (donor-robust AND co-detection, with donor and cell support):

tissue	cell_type	cover- age	n_ - donors	n_cells
respiratory sys- tem	mucus secreting cell	0.167	5	269
small intestine	paneth cell of epithelium of small intestine	0.143	5	1657
tongue	salivary gland cell	0.14	2	734
small intestine	intestinal crypt stem cell of small intestine	0.095	5	531
lung	respiratory goblet cell	0.071	4	978
respiratory sys- tem	basal cell	0.065	5	10003

Uncertainty (bootstrap over patients; 18 patients, 2000 resamples). Resampling the unit of replication, patients rather than cells, gives between-subject intervals for PSMA x STEAP1:

- Median coverage **0.69** (95% CI 0.58–0.79); the per-patient floor **Q0.10 = 0.45** is much less stable (95% CI 0.19–0.61) and is reported as indicative, not precise. Coverage is at least 0.4 in 89% of patients.

- The pair stays **Pareto-non-dominated in 99.6%** of patient resamples, so its frontier status is not an artifact of a few patients.
- Worst extra-prostatic liability (respiratory system | mucus secreting cell): donor median **0.17**, but the bootstrap **95% upper bound is 0.37** over 5 donors. A conservative safety read should use the upper bound, which is more than double the median.
- Malignant-minus-benign co-detection, paired within patient: median **0.54** (95% CI 0.48–0.60), positive in **100%** of 18 patients with both compartments, so the tumor enrichment is not a compositional average.

De-leaked sensitivity (label leakage). Recomputing the Liu and Wallace signatures with all 29 panel genes removed barely changes them (Spearman **0.99** against the full signature within malignant cells). A leakage-free malignant proxy, CopyKAT-aneuploid cells top-ranked by the de-leaked signature and count-matched, recovers **82%** of the authors’ malignant cells, and PSMA x STEAP1 coverage on it is essentially unchanged (**0.63** vs 0.67). The clean lead is not an artifact of the panel genes helping to define the compartment. This is a partial control, not the authors’ full reannotation.

Do the two markers escape on different healthy cells? An AND gate only lowers liability if PSMA and STEAP1 are positive on different normal cells. At each marker’s worst single tissue the other is nearly absent, so the gate collapses it: PSMA’s duodenal liability (0.85) falls to 0.01 because STEAP1 is off there, and STEAP1’s airway liability (0.50) falls to 0.07 because PSMA is off there. But the two single liabilities are moderately correlated across populations (Spearman **0.51**), and the residual AND liability sits in secretory epithelium (mucus, paneth, salivary) where the markers partly co-escape (co-escape ratio = share of the limiting marker’s positive cells that are double-positive, up to ~0.9). The safety gain is real at the extremes but partial where the two escapes overlap:

population	PSMA	STEAP1	AND	co-es- cape	donors
respiratory system	mucus secreting cell	0.41	0.25	0.17	0.67
small intestine	paneth cell of epithelium of small intestine	0.83	0.16	0.14	0.9

population	PSMA STEAP1 AND	co-es- cape	donors
tongue	salivary gland cell	0.39 0.33	0.14 0.42
small intestine	intestinal crypt stem cell of small intestine	0.43 0.14	0.1 0.67
lung	respiratory goblet cell	0.1 0.5	0.07 0.73
respiratory system	basal cell	0.11 0.34	0.07 0.58

4.5 NOT gates: a negative result

No NOT gate (activator on the tumor, blocker sparing a normal tissue) achieved both high malignant coverage and low predicted normal-tissue escape. The 10 best-ranked exploratory activator/blocker combinations still retain a large activator-positive / blocker-negative fraction in normal tissue (worst extra-prostatic co-detection shown, with donor support), so none is an attractive candidate; the blocker-negative call also depends on trusting scRNA dropout. This is reported as a negative result:

activator	blocker	Q0.10	median	worst xp	worst cell type	donors
STEAP1	SLC34A1	0.739	0.874	0.496	respiratory goblet cell	4
STEAP1	CDH16	0.739	0.874	0.496	respiratory goblet cell	4
STEAP2	MUC1	0.6	0.843	0.477	fibroblast	5
STEAP1	MUC1	0.543	0.732	0.411	fibroblast	5
STEAP2	SLC34A1	0.936	0.968	0.653	bladder urothelial cell	6

activator	blocker	Q0.10	median	worst xp	worst cell type	donors
STEAP2	CDH16	0.936	0.968	0.656	bladder urothelial cell	6
TM-PRSS2	CDH16	0.894	0.97	0.903	enterocyte of epithelium of large intestine	3
TM-PRSS2	SLC34A1	0.894	0.97	0.905	enterocyte of epithelium of large intestine	3
CD46	SLC34A1	0.856	0.958	0.963	bladder urothelial cell	6
CD46	CDH16	0.856	0.958	0.965	bladder urothelial cell	6

5 Replication in an independent, advanced-disease cohort

We replicated the nominated pair in **HuPSA** [10], an independent prostate-cancer meta-analysis of **368,831 cells** from **6 10x studies** spanning normal, localized, castration-resistant, metastatic (mCRPC) and neuroendocrine disease. Malignancy uses HuPSA’s own cell-state labels (AdPCa, NEPCa, KRT7, progenitor-like), independent of our Liu/Wallace + CopyKAT call, so this is also a label-independent check. Replication is on tumor coverage; the healthy-liability side stays with Tabula Sapiens. Two findings:

- **Tumor-specificity replicates.** Malignant-minus-benign-epithelial AND co-detection is **+0.15**, positive in **83%** of 12 paired samples.
- **Coverage does not generalize to advanced disease.** Median per-patient coverage across HuPSA is **0.15** (vs 0.69 in the localized cohort). It is AR-lineage-dependent: **0.37** in AR+ adenocarcinoma but essentially **0.00** in the neuroendocrine, KRT7 and progenitor-like states, and it falls from **0.18** in localized disease to **0.05** in mCRPC. The pair marks AR-driven adenocarcinoma and is lost where lineage plasticity takes over, exactly the metastatic and neuroendocrine population most cell-therapy development targets.

PSMA x STEAP1 AND co-detection by malignant cell state (HuPSA):

malignant state	AND coverage	n cells
AdPCa_AR+_2	0.37	2638
AdPCa_AR+_1	0.35	27051
AdPCa_Proliferating	0.17	460
AdPCa_AR+_3	0.13	198
AdPCa_ARlo	0.07	5957
AdPCa_ARhi	0.06	3904
NEPCa	0	8081
Progenitor_like	0	5993

malignant state	AND coverage	n cells
KRT7	0	1937

6 Limitations

- **Label leakage from the malignant annotation.** The discovery cohort defines malignant cells partly from prostate-cancer expression signatures (Liu, Wallace) that contain HPN and EPCAM. Pairs with those genes are excluded as confounded, and a de-leaked sensitivity (Robustness) shows the clean lead survives a malignant compartment defined without the panel genes. That check is partial; a complete fix would recompute the authors' annotation with all screened genes removed and a second CNV method.
- **Transcript co-detection is a selectivity surrogate, not therapeutic safety.** scRNA measures mRNA, not surface antigen density, extracellular epitope availability, binding affinity, shedding, internalization, the minimum antigen density for activation, or residual single-arm signalling. HPA localization supports membrane-localization plausibility but does not qualify targetability; same-cell surface co-expression needs CITE-seq, multiplex IHC, or dual-color flow to confirm.
- **Statistical uncertainty is quantified for the lead, not the whole table.** The nominated pair now carries bootstrap-over-patient intervals for coverage, a Pareto-membership probability, a donor-level 95% upper bound for its worst liability, and a paired malignant-benign delta. A full hierarchical (beta-binomial mixed) model over all 325 pairs, with the selection re-run inside each bootstrap, remains future work, so the ranking of the non-lead pairs is still point estimates.
- **Assay-matched but single-assay.** The healthy reference is restricted to 10x 3' v3 to match the tumor, so Smart-seq2 and other-chemistry cells in Tabula Sapiens are excluded and a liability seen only on another platform would be missed. Dissociation also selectively loses fragile populations.
- **NOT-gate calls trust B-negatives**, which are dropout-sensitive, and none cleared the liability bar (a negative result).
- **Advanced disease.** The pair is nominated on localised hormone-naive cancer. Replication in HuPSA shows its coverage holds in AR+ adenocarcinoma but is largely lost in castration-resistant, metastatic and neuroendocrine disease, so it does not currently address the population most cell therapies target.
- **Malignant labels are the authors' CNV/signature calls**, adopted rather than re-derived here.
- **The PSMA-PSCA benchmark is preclinical.** It is a safety/selectivity benchmark, not a coverage benchmark; its low co-detection is consistent with limited PSCA transcript detection in this dataset.

7 Conclusion

This analysis is a patient-aware, transcriptome-based prioritization of candidate combinatorial antigens in localized prostate cancer. The PSMA-PSCA benchmark shows the expected collapse in normal-tissue transcript co-detection relative to its single markers. A single clean pair is nom-

inated, PSMA x STEAP1: higher malignant co-detection than the benchmark and, once the leakage-confounded HPN pairs are removed, Pareto-non-dominated among the surface-accessible candidates. The target pair is already in clinical development as an OR-like dual antibody-drug conjugate (ABBV-969), so the contribution is the AND framing, not the pair. This is a hypothesis-generating ranking, not evidence of therapeutic safety. Replication in an independent cohort (HuPSA) confirms the pair’s tumor-specificity but shows its coverage is confined to AR-driven adenocarcinoma and largely lost in metastatic and neuroendocrine disease, so it marks localised AR+ cancer rather than the advanced population most therapies target. Protein-level same-cell measurement (CITE-seq, multiplex IHC) and recomputed leakage-free labels remain required before it can be called a validated target.

8 Reproducibility

The pipeline runs from a clean clone with `uv`; every table in `results/tables/` is regenerated by the numbered scripts in `scripts/`, and this report redraws its figures from those tables. See the README for the exact command sequence.

9 References

Bibliography

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